

MECHTRON/SFWR ENG 3DX4 Tutorial Quiz 6 L03: Statespace to Transfer Function & Back

1. (10 marks)

a) (7 marks) Find a statespace representation of the following system:

$$G(s) = \frac{30}{s^5 + 8s^4 + 9s^3 + 6s^2 + s + 30}$$

b) (3 marks) What formula would you compute to obtain the transfer function for your state space system you created in part 1a? NOTE: You do not have to evaluate this formula to get full marks, but you can if you want to check part 1a! Evaluate the formula for your answer to part (a) above using the Matlab symbolic toolbox (you don't need to submit the matlab code).